

GENIE for FASER

A native build without the ATLAS container,
and what changes from 3.04 to 3.06

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1. Why the old recipe needed the ATLAS container
2. The new build: one repository, two scripts
3. Pitfalls solved along the way (macOS / modern ROOT)
4. GENIE 3.04 (FASER fork) → 3.06.02: what changed
5. Validation: event rate and a geometry-unit bug
6. Next steps

The old recipe

How 3.04 was built

- `setupATLAS -c centos7`: Apptainer container with CentOS 7
- `asetup Athena,22.0.49`: only to get compilers and libraries from CVMFS (gcc 11, ROOT 6.24, Pythia6, LHAPDF, GSL, ...)
- TPythia6 built as an *Athena/Calypso* CMake project
- GENIE itself: plain `./configure && make`

Problems

- CentOS 7 is end-of-life: the container exists only to run old binaries
- mixed gcc8 and gcc11 libraries on one `LD_LIBRARY_PATH`
- broken Pythia6 link line in `Make.include`
- not usable on a Mac

Key point

GENIE contains no ATLAS code. Athena was only a convenient way to get a software stack.

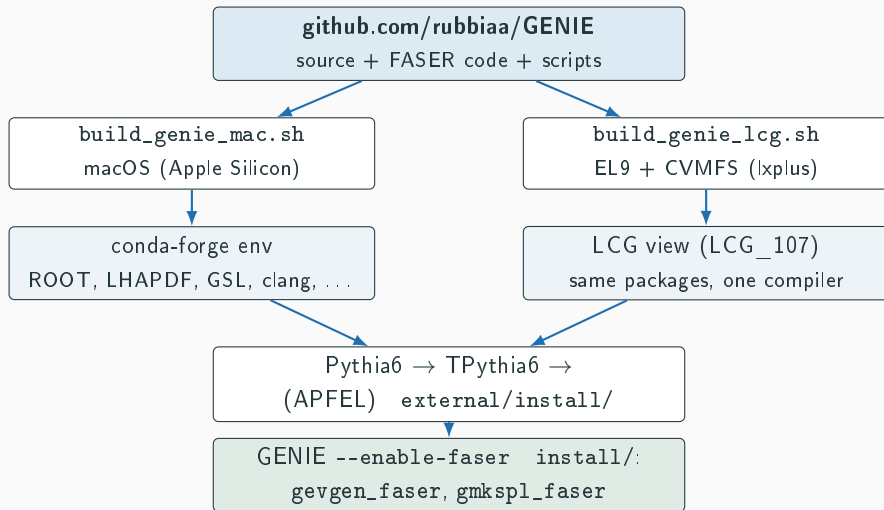
What GENIE actually needs

Dependency	Role	Source in the new build
ROOT (with Geom)	I/O, geometry (TGeo), maths	conda-forge / LCG view
Pythia 6.4.28	hadronization, decays	built from alisw/pythia6
TPythia6 (libEGPythia6)	ROOT interface to Pythia6	built here (plain CMake)
LHAPDF 6	PDFs	conda-forge / LCG
GSL, libxml2, log4cpp	numerics, config, logging	conda-forge / LCG
<i>optional</i> : APFEL 3.0.6	HEDIS structure functions	WITH_APFEL=1
<i>optional</i> : Pythia 8	Pythia8 hadronization	WITH_PYTHIA8=1

TPythia6 was removed from ROOT in 6.32, so it is compiled from the old ROOT sources.

APFEL is off by default: the FASER splines and tunes do not use HEDIS.

The new build



No container, no Athena, no asetup. The same scripts work on both platforms.

Repository = work directory

```
GENIE3.06/                <- git repository
|-- README.md
|-- GENIE_3.04_vs_3.06.md
|-- setup_mac.sh    build_genie_mac.sh
|-- setup_lcg.sh    build_genie_lcg.sh
|-- GenieGenerator/    <- GENIE 3.06.02 + FASER
|  |-- src/    config/    data/
|  |-- faser/    (geometries, fluxes, scripts)
|  |  |-- TPythia6_standalone/
|  |  `-- build/external/    (patch, Pythia6 CMake)
|-- install/          \
|-- external/          |    created by the
|-- build/              |    build, ignored by git
|-- faser_xsec/          |
`-- run/                /
```

- Full GENIE history kept (7 962 commits), FASER work on top
- Build products, downloads, splines and event files are *git-ignored*
- Large Kling 2023 flux ntuples (≤ 575 MB) not in git: regenerated with `getRawFluxes.sh` + `convertAllFluxes.sh`
- Scripts find the source tree and work directory automatically

Building on a Mac

```
git clone https://github.com/rubbiaa/GENIE.git GENIE3.06
cd GENIE3.06
./build_genie_mac.sh          # first time ~20 min
source setup_mac.sh           # every new terminal (zsh or bash)
```

What the script does:

1. installs a private Miniforge in ~/miniforge3 (does not touch .zshrc or an existing Anaconda)
2. creates the genie env: root lhpdf gsl libxml2 log4cpp cmake make clang gfortran
3. builds Pythia 6.4.28 and TPythia6 into external/install/
4. patches Make.include for Apple Silicon and the conda compilers (idempotent)
5. ./configure --enable-faser ..., make, make install

Options: WITH_PYTHIA8=1, WITH_APFEL=1, NJ=<jobs>.

Building on lxplus (EL9 + CVMFS)

```
git clone https://github.com/rubbiaa/GENIE.git GENIE3.06 && cd GENIE3.06
./build_genie_lcg.sh                # LCG_107, x86_64-el9-gcc13-opt
LCG_VERSION=LCG_105 ./build_genie_lcg.sh # any other release
source setup_lcg.sh
```

- One LCG view provides ROOT, Pythia6, LHAPDF, GSL, log4cpp and libxml2, all built with the same compiler
- `setup_lcg.sh` replaces `ATLAS_container.sh`, `.asetup.save`, `kk` and `setupGenerator.sh`
- Only TPythia6 and GENIE are compiled locally

Pitfalls solved

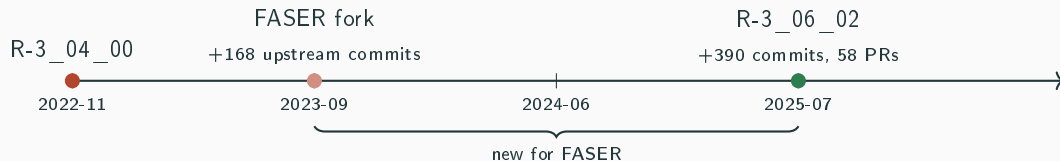
Problems found and fixed (all automatic in the scripts)

Symptom	Cause → fix
conda solve never finishes expected ')' in <charconv>	old Anaconda classic solver → private Miniforge with mamba clang 18 with libc++ 20 headers (ROOT pin) → match clang to the headers
undeclared identifier NAN	macOS 27 SDK takes NAN from <float.h>; strict -std=c++20 hides it → -std=gnu++20
libc++ “feature unavailable” rootcling fails	conda targets macOS 11 → -D_LIBCPP_DISABLE_AVAILABILITY dictionary step does not see CXXFLAGS → flags also passed via the include options
-lPythia6/-lEGPythia6 not found	make distclean empties install/ → externals in external/install/; library path added
APFEL: arm64-apple unknown Apple Silicon flags	2013 config.sub → use aarch64-apple-darwin generic macosx64 block (Xcode-10 era) → dedicated macosxarm64 block

3.04 → 3.06



Where the versions sit



- R-3_06_02 in the FASER GitLab is a plain mirror of the official release: **no FASER code**
- Default tune unchanged: G18_02a_00_000; its model configuration is identical
- About 1 480 files differ upstream (configs, new models, data tables)

FASER code ported to 3.06

Item	Status
gevgen_faser, gmkspl_faser	ported unchanged
FaserGMCJDriver, FaserROOTGeomAnalyzer	ported; base interfaces unchanged
configure --enable-faser, Apps/Makefile	applied cleanly
GVLD-Emax = 7 TeV (official: 1 TeV)	re-applied
NaturalIsotopes::GetIsotopeData()	re-added
weighted-flux POT fix in GSimpleNtpFlux	already upstream
tunes F23_00a, G25_01a	ported
faser/ (geometries, fluxes, scripts)	copied; big fluxes git-ignored
TGeo length units (see validation)	new fix

Not ported: the fork's broken Pythia6 lines in `Make.include`, which also removed the Pythia8 link block.

Upstream changes relevant to FASER

Pythia6 / Pythia8

- Pythia6 now optional (`--disable-pythia6`)
- decayer: `PythiaDecayer` → `Pythia6Decayer2023` (and `Pythia8Decayer2023`)
- DIS charm: `AGCharm2019` → `AGCharmPythia6Hadro2023` (same parameters)
- HEDIS hadronization: `LeptoHadPythia6/8`
- HE leptons: separate Pythia6/8 W-decay classes

Other

- memory-leak fixes (long productions)
- G_F , α configurable in `CommonParam.xml`
- build: Apple Silicon support, no ROOTSYS needed

Not used by default

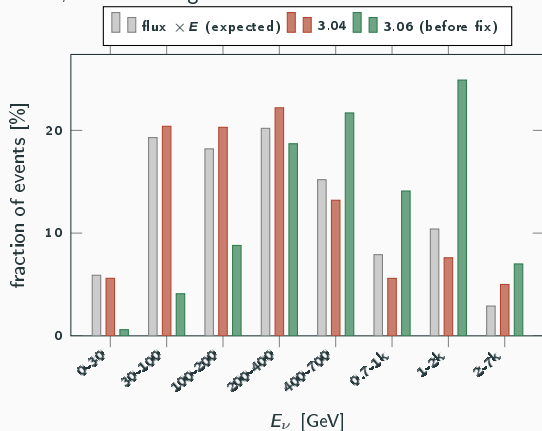
- tunes G24_20i/j/k/l, N24_20i, MK19_00a
- MK single-pion model, Z-expansion and MARun form factors, SuSAv2/CRPA updates

⇒ Expect small event-level differences in decays and charm hadronization: validate before production.

Validation

First comparison: 774 vs 40 085 events

Same command, flux, geometry and (3.04) splines: `gevgen_faser -l 1000 (1000 fb-1)`, FASERCal V10, Aki 2024 light flux.



- 3.04 follows $\text{flux} \times \sigma \propto E$, as expected for DIS
- 3.06 deficit depends on energy: $\times 450$ below 30 GeV, $\times 16$ above 1 TeV
- Splines, flux code and FASER code were checked: **not** the cause

Cause: TGeo length units

	3.04	3.06 (before fix)	true geometry (GDML)
vertex x (5–95%)	0.4–1.2 m	0.04–0.14 m	detector centred at 0.45 m
vertex z (5–95%)	0.9–6.7 m	0.04–0.68 m	target 0–2.9 m, MDT at 6.1 m

- The FASER apps hard-wire geometry lengths in **mm**
 - ROOT 6.16–6.24 (3.04 production): TGeo default `kG4Units` (mm) ✓
 - ROOT $\geq$ 6.26 (6.40 here): default `kRootUnits` (cm)
- ⇒ detector $10\times$ too small: shorter paths, and only the forward (high-energy) neutrinos hit it

```
// FaserROOTGeomAnalyzer::Load()
TGeoManager::LockDefaultUnits(false);
TGeoManager::SetDefaultUnits(
    TGeoManager::kG4Units); // mm
TGeoManager::LockDefaultUnits(true);
TGeoManager::Import(filename);
```

Status

Fixed and committed. The rerun should give $\approx$ 40k events, compatible with 3.04.

Next steps

1. **Confirm the units fix:** rerun the 1000 fb^{-1} light and charm samples and compare with 3.04 (rates, E_ν , targets, vertices)
2. **Regenerate splines for 3.06** with `faser/Splines/` (free nucleons $\rightarrow$ nuclei, 6 flavours, up to 7 TeV; $\sim 2\text{--}3$ days on 16 cores)
3. **Physics validation** of the rewritten decays and charm hadronization (charm and τ samples)
4. **Optional:** Pythia8 hadronization (`WITH_PYTHIA8=1`), then drop Pythia6
5. **Share:** push `faser-R-3_06_02` back to the FASER GitLab

Backup: command cheat sheet

```
# environment
source setup_mac.sh                # or setup_lcg.sh on lxplus

# incremental rebuild after a code change
cd GenieGenerator && make -j && make install && cd ..

# event generation (FASERCal V10, Aki 2024 light flux, 1000 fb-1)
gevgen_faser -l 1000.0 -r 0 -g $GENIE/faser/FASERCAL_V10.gdml \
  -f $GENIE/faser/Fluxes/Aki_2024/events_light_4x4.root --seed 2999833 \
  --cross-sections $GENIE_HOME/faser_xsec/faserSplines.7TeV.xml \
  -o fasercal.Aki2024.v10.light

# splines (run in GENIE3.06/splines_work, one flavour at a time)
source $GENIE/faser/Splines/calcSplinesN.sh; wait
source $GENIE/faser/Splines/mergeSplinesN.sh
source $GENIE/faser/Splines/calcNuMuSplinesA.sh; wait      # ... all 6 flavours
source $GENIE/faser/Splines/mergeSplinesA.sh
```